$\mathcal{F}(S) := \{ f : S \longrightarrow \mathbb{R} \}$

$A \subseteq S$

$$I_A = \left\{ f : S \to \mathbb{R} \;\middle|\; f(a) = 0 \ \forall a \in A \right\}$$

is an ideal of $\mathcal{F}(S)$.

If A is singleton let's say $A = \{a\}$.

Then $I_A = \{ f : S \to \mathbb{R} \mid f(a) = 0 \}$.

Now, let $g, h \in \mathcal{F}(S)$ &

$$
\begin{aligned}
gh \in I_A \Rightarrow\ & (gh)(a) = 0 \\
\Rightarrow\ & g(a)h(a) = 0 \\
\Rightarrow\ & g(a) = 0 \text{ or } h(a) = 0 \\
\Rightarrow\ & g \in I_A \text{ or } h \in I_A.
\end{aligned}
$$

$\Rightarrow I_A$ is a prime ideal.

$C[a,b] = \{ f : [a,b] \to \mathbb{R} \mid f \text{ is cts} \}$

$C[a,b] \leq \mathcal{F}([a,b])$

Existence of a Maximal Ideal:

Let $\sum = \{ I : I \underset{\neq}{\triangleleft} A \}$; $A \neq \langle 0 \rangle$

$\langle 0 \rangle \in \sum \Rightarrow \sum \neq \varphi$.

(*)

Let $I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$ be a chain in

$\bigcup_{n=1}^{\infty} I_n$ is an ideal

If $\bigcup_{n=1}^{\infty} I_n = A \Rightarrow 1 \in I_n$ for some $n \in \mathbb{N}$.

$\Rightarrow I_n = A$ as $I_n \in \sum$

$\Rightarrow \bigcup_{n=1}^{\infty} I_n \in \sum$, and this is an upper bound for the chain (*).

Then by Zorns Lemma, $\sum$ has a maximal element say $\mathfrak{M}$. If $\mathfrak{M}$ is not a maximal element $\exists$ a proper ideal $J \underset{\neq}{\subseteq} A$ and

$\mathfrak{M} \underset{\neq}{\subseteq} J \underset{\neq}{\subseteq} A$ but $J \underset{\neq}{\subseteq} A \Rightarrow J \in \sum$

this contradicts $\mathfrak{M}$ is maximal element of $\sum$.

$\underline{\underline{Th^m}}$: If A is a non-zero commutative ring with identity, then A has a maximal ideal.

needed in finding upper bound

$\underline{\underline{\text{Cor}}}$: If A is a commutative ring with identity & I is a proper ideal of A, then $\exists$ a maximal ideal containing I.

$\underline{\underline{\text{Pf}}}$: Take $I \subsetneq A$ & consider A/I.

Now, A/I is CRI. $\Rightarrow$ A/I has a maximal ideal m. Then we have $\varphi : A \longrightarrow A/I$

$$a \longmapsto a + I$$

so $\varphi^{-1}(m) \supseteq I$.

And as $m \neq A/I \Rightarrow \varphi^{-1}(m) \neq A$.

& $I \subsetneq \varphi^{-1}(m) \subsetneq A$ [$\because m \neq \langle 0 \rangle$ $\Rightarrow I \subsetneq \varphi^{-1}(m)$]

Now, if $J \subsetneq A$ s.t.

$$I \subsetneq \varphi^{-1}(m) \subsetneq J \subsetneq A.$$

$$m \subsetneq J/I \subsetneq A/I \quad \times$$

φ^{-1}

$\varphi(\varphi^{-1}(A)) = A$ as φ is onto

Cor: If $x \in A$ & $x \neq 1_A$ then $\exists\, m \subsetneq A$ s.t. $x \in m$ and m is maximal ideal of A.

Let $I = \{x \in A : x \text{ is a non-unit}\}$

Suppose $I \trianglelefteq A \Rightarrow I$ is a maximal ideal. and any $x \in I$ we have I is the only maximal ideal containing x and infact I is the only maximal ideal possible as proper ideal must contain all non-units.

Prop: If the set of all units form an ideal of A, then I is the unique maximal ideal in A.

Defn: A ring A is said to be a local ring if it has only one maximal ideal.

Example: ① Fields.

② $\mathbb{Z}_{p^n}$ where p is a prime.

Prop: Let A be a CRI and $\mathfrak{m}$ be a maximal ideal. If $1+x \in \mathfrak{m}$ is a unit $\forall x \in \mathfrak{m}$, then $\mathfrak{m}$ is the unique maximal ideal of A.

$$(\mathfrak{m} \underset{\neq}{\trianglelefteq}! A)$$

Pf: Let x be a non-unit. If $x \notin \mathfrak{m}$, then $\langle \mathfrak{m}, x \rangle = A \Rightarrow \exists\, u \in \mathfrak{m}$ & $r \in A$ s.t.

$$1 = u + rx \Rightarrow 1 - u = rx \Rightarrow rx \text{ is unit}$$

$\Rightarrow x$ is a unit $(\Leftarrow\!\!\Rightarrow\!\Leftarrow)$

Hence $x \in \mathfrak{m} \Rightarrow \boxed{\mathfrak{m} \underset{\neq}{\trianglelefteq}! A.}$

Let $\mathfrak{N} = \{x \in A : x \text{ is a nilpotent}\}$

If $x, y \in \mathfrak{N}$, $\Rightarrow x^n = 0$ & $y^m = 0$ for some $n, m \in \mathbb{N}$

$\Rightarrow (x+y)^{n+m} = 0. \Rightarrow x+y \in \mathfrak{N}$

If $x \in \mathfrak{N}$, $-x \in \mathfrak{N}$

If $x \in \mathfrak{N}$ & $r \in A \Rightarrow rx \in \mathfrak{N}$?

Prop: Set of all nilpotent elements form an ideal of A.

If $x \in \mathfrak{N}$, then $x^n = 0$.

Suppose $\mathfrak{p}$ is a prime ideal of A. If $x \in \mathfrak{N}$ then $x^n = 0 \in \mathfrak{p}$ for some $n \in \mathbb{N} \Rightarrow x \in \mathfrak{p}$.

$$\Rightarrow \mathfrak{N} \subseteq \mathfrak{p} \quad \forall\, \mathfrak{p} \text{ prime ideals}$$

$$\Rightarrow \mathfrak{N} \subseteq \bigcap_{\substack{\mathfrak{p} \trianglelefteq A \\ \mathfrak{p} \text{ is prime}}} \mathfrak{p}$$

$$\Sigma = \left\{ I \underset{\neq}{\trianglelefteq} A \;\middle|\; x^n \notin I \;\; \forall\, n \in \mathbb{N} \right\}$$

$\langle 0 \rangle \in \Sigma \Rightarrow \Sigma \neq \phi$.

Let $I_1 \subseteq I_2 \subseteq \cdots$ be a chain in Σ

Let $I = \bigcup_{n \in \mathbb{N}} I_n \underset{\neq}{\trianglelefteq} A$.

$$\Rightarrow I \in \Sigma$$

By Zorn's Lemma, $\exists\, \mathfrak{J}$ be a maximal elt. of Σ.

Claim: $\mathfrak{J}$ be a prime ideal.

Let $a, b \in$ ま, then if $a \notin$ ま & $b \notin$ ま then

$\text{ま} + (a) \notin \Sigma$ and $\text{ま} + (b) \notin \Sigma$

$\Rightarrow \exists\, n$ & m with

$x^n \in \text{ま} + (a)$ & $x^m \in \text{ま} + (b)$

$\Rightarrow x^n = z_1 + r_1 a$ & $x^m = z_2 + r_2 b$

$x^{n+m} = z_1 z_2 + z_1 r_2 b + z_2 r_1 a + r_1 r_2 ab \in \text{ま}.$

$\Rightarrow \text{ま} \notin \Sigma$ (contradiction!!)

$\Rightarrow$ ま is a prime ideal & $x^n \notin \text{ま} \ \forall n \in \mathbb{N}$

$\Rightarrow x \notin \text{ま}.$

$$\Rightarrow x \notin \bigcap_{\substack{p \trianglelefteq_{\neq} A \\ p \text{ is prime}}} p$$